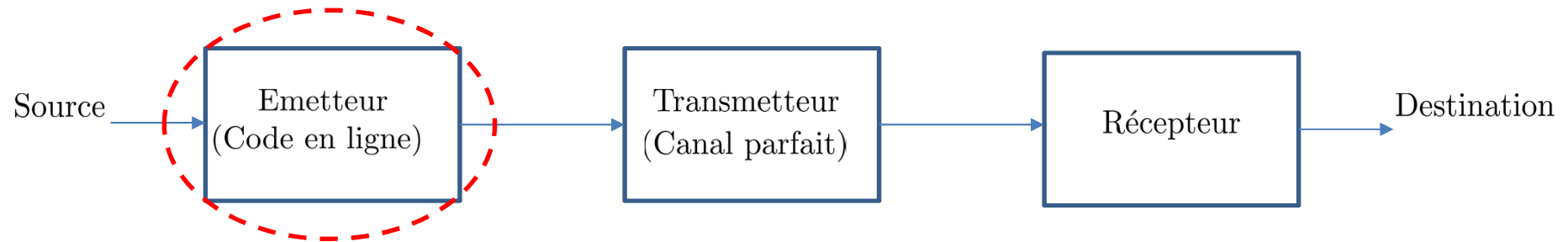


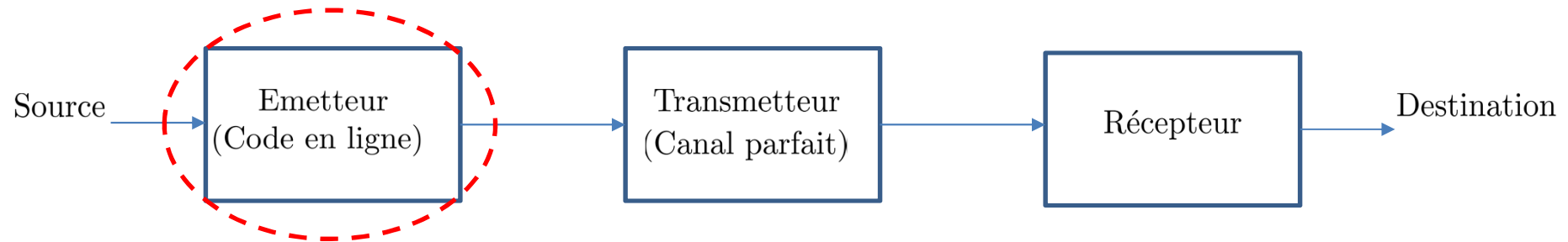
Séance 2



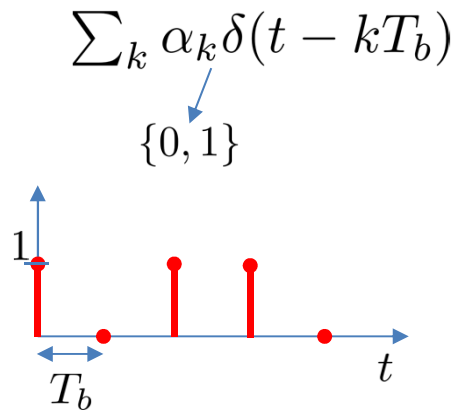
Code en ligne (modélisation)



Séance 2

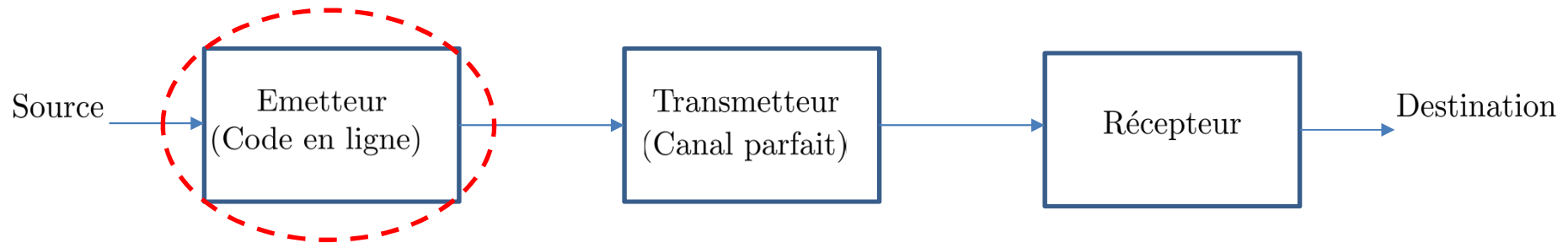


Code en ligne (modélisation)



Débit binaire : $D = \frac{1}{T_b}$ bits/s

Séance 2

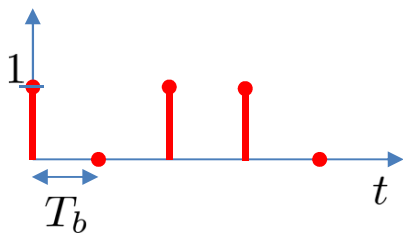


Code en ligne (modélisation)



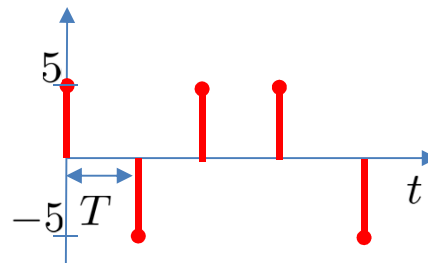
$$\sum_k \alpha_k \delta(t - kT_b)$$

$\{0, 1\}$



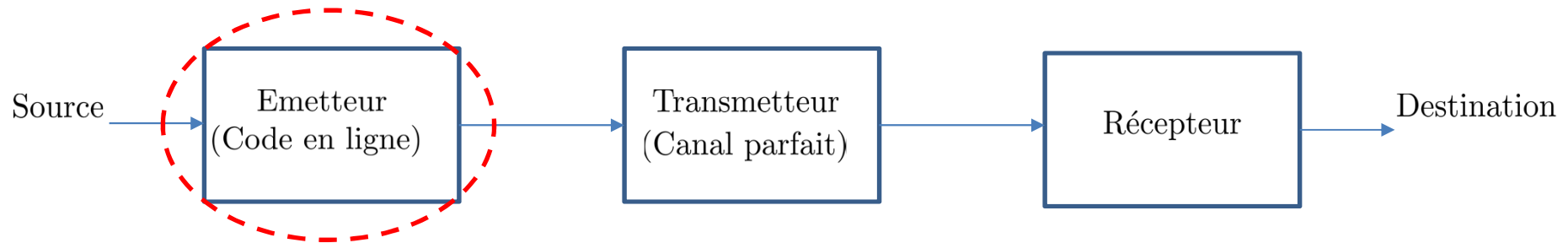
Débit binaire : $D = \frac{1}{T_b}$ bits/s

$$\sum_k a_k \delta(t - kT)$$



Rapidity de modulation : $R = \frac{1}{T}$ bauds

Séance 2

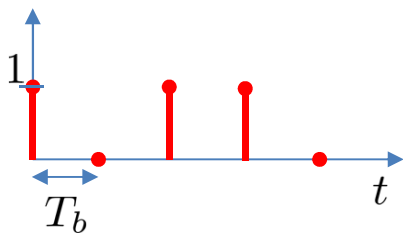


Code en ligne (modélisation)



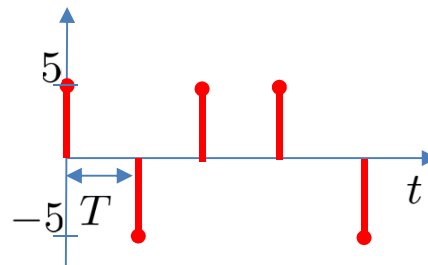
$$\sum_k \alpha_k \delta(t - kT_b)$$

$\{0, 1\}$



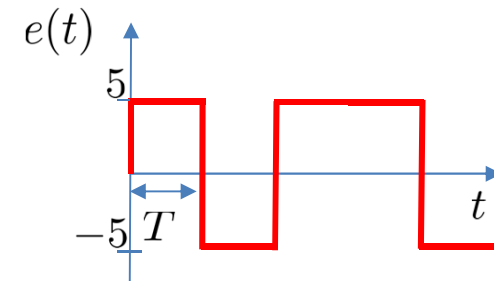
Débit binaire : $D = \frac{1}{T_b}$ bits/s

$$\sum_k a_k \delta(t - kT)$$



Rapidity de modulation : $R = \frac{1}{T}$ bauds

$$e(t) = \sum_k a_k h(t - kT)$$

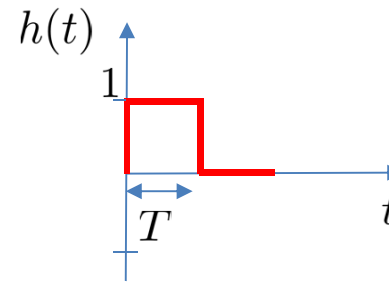


Séance 2

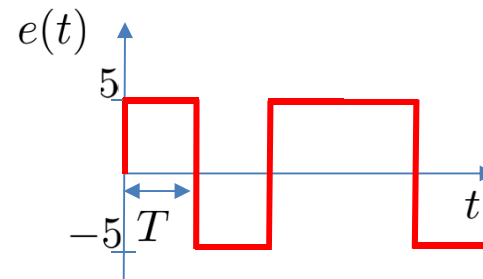
3 codes en ligne

1-NRZ binaire

$$\begin{cases} \alpha_k = 1 \rightarrow a_k = A_{\max} \\ \alpha_k = 0 \rightarrow a_k = A_{\min} \end{cases}$$



Exemple : $A_{\max} = 5$, $A_{\min} = -5$



PC = monde numérique
⇒ Comment simuler un signal analogique ?

Séance 2

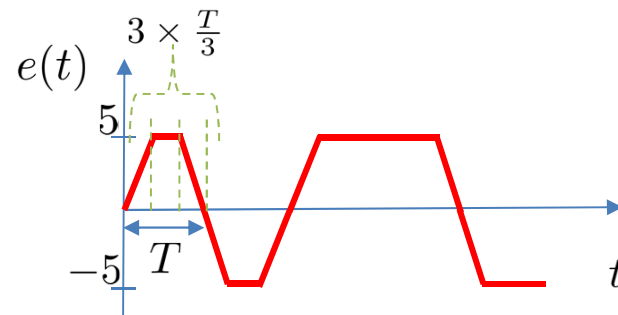
3 codes en ligne

2-NRZT binaire

$$\begin{cases} \alpha_k = 1 \rightarrow a_k = A_{\max} \\ \alpha_k = 0 \rightarrow a_k = A_{\min} \end{cases}$$

$h(t)$ change en fonction de la valeur des symboles précédents et suivants
 $\Rightarrow$ Simulation des temps de montée/descente des composants électroniques

Exemple : $A_{\max} = 5$, $A_{\min} = -5$

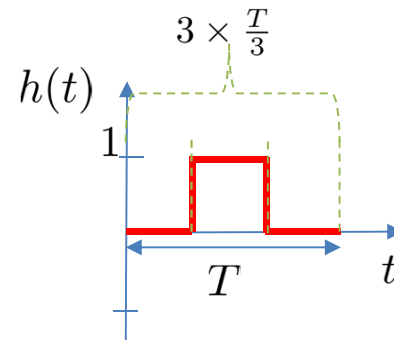


Séance 2

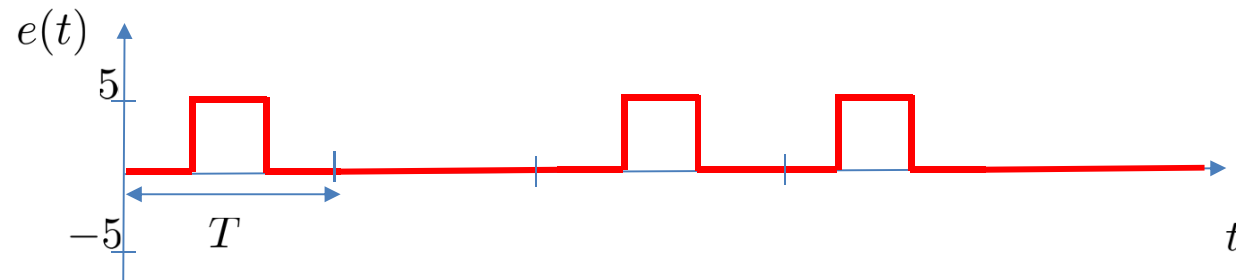
3 codes en ligne

3-RZ binaire

$$\begin{cases} \alpha_k = 1 \rightarrow a_k = A_{\max} \\ \alpha_k = 0 \rightarrow a_k = A_{\min} \end{cases}$$



Exemple : $A_{\max} = 5$, $A_{\min} = 0$



Séance 2

Contraintes

NRZ et NRZT

$$\left\{ \begin{array}{l} A_{\max} \geq 0 \\ A_{\min} \leq 0 \\ A_{\min} < A_{\max} \end{array} \right.$$

RZ

$$\left\{ \begin{array}{l} A_{\max} \geq 0 \\ A_{\min} = 0 \\ A_{\min} < A_{\max} \end{array} \right.$$